

Toronto, ON
(437)-421-1277

Ruth Zhang

linkedin.com/in/ruthxzhang/
ruthxzhang.github.io

ruthruth.zhang@mail.utoronto.ca

EDUCATION

- University of Toronto**
Master of Science in Applied Computing (Applied Mathematics Concentration)
- Toronto, ON
Sept 2026 –
- University of Toronto**
BSc Mathematical Applications in Economics & Finance Specialist, Statistics Minor
- Toronto, ON
Sept 2022 – Apr 2026
- **GPA: 3.98/4.0**; Dean's List (2023, 2024, 2025)
 - **Relevant Coursework:** Machine Learning, Programming in Python & R, Probability & Statistics, Data Analysis, Mathematical Logic, Real Analysis, Differential Equations, Multivariable Calculus, Linear Algebra, Optimization
 - **Awards:** University of Toronto Scholar, AI And Eleanor May Scholarship, Innis College Alumni & Friends Award

SKILLS

Programming Languages: Python, R, JAVA, SQL
Technologies: MySQL, Git, GitHub, HTML/CSS, $\text{\LaTeX}$, Microsoft Excel, Airtable

RESEARCH EXPERIENCE

- Research Assistant | University of Toronto**
Supervisor: Avi Goldfarb (Rotman Chair in Artificial Intelligence & Healthcare)
- Toronto, ON
Jan 2025 – May 2026
- Illusion of Competition Project*** (Sept 2025 – Present)
- **Data Engineering & Analysis:** Designed and implemented reproducible R workflows (version-controlled with Git) to integrate and analyze large-scale consumer datasets (more than 1,000,000 observations), uncovering brand ownership structures and market dynamics across multiple product categories.
 - **Brand Interaction Modeling:** Built conditional dependence measures across consumer consideration sets to quantify directional brand-brand relationships, enabling empirical tests of independence assumptions.
 - **Data Pipeline Development:** Consolidated and cleaned disparate survey and experimental data into a unified analysis-ready dataset, enabling empirical research on consumer misperception of competition.
- Lab for the Management of Science and Technology*** (May 2025 – Aug 2025)
- **Web Scraping Pipeline:** Developed Python scripts to automate extraction, cleaning, and standardization of 15+ quarterly financial variables for four major semiconductor firms (TSMC, Intel, AMD, NVIDIA) from web and API sources, producing publication-ready Excel datasets for research.
 - **Exploratory Data Analysis:** Performed EDA in R on a dataset of 5,000+ procurement records to identify key spending patterns, providing data-driven insights for the Acceleration Consortium's budget allocation.
 - **Computer Vision Data Preparation:** Executed detailed visual classification of over 1,500 retail storefront images, facilitating empirical research on the correlation between storefront design and passerby foot traffic.
- Creative Destruction Lab*** (Jan 2025 – Apr 2025)
- **Machine Learning Validation:** Validated the performance of a zero-shot classification algorithm (GPT-o3-mini) by manually labeling a stratified random sample of 500 open-text startup objectives, achieving an 81% agreement rate with the model and confirming its reliability for automated coding tasks.
 - **Data Curation & Linkage:** Cross-referenced and validated over 300 funding round records for early-stage, science-based startups, ensuring accurate linkage to PitchBook, an external database.

WORK EXPERIENCE

- Teaching Assistant | University of Toronto**
Courses: MAT135 – Calculus I, MAT136 – Calculus II
- Toronto, ON
Sept 2024 – Apr 2025
- Conducted weekly tutorials for classes of 30 students, clarifying complex concepts in Calculus.
 - Collaborated with course instructors to design and implement strategies for improving student participation.